UNIVERSITI TEKNOLOGI PETRONAS | INTERNATIONAL FINANCIAL MANAGEMENT

FinBridge AI / Finance Tutor + Scenario Engine

Turn international finance theory into an interactive decision lab for non-finance students.

CLO3 · 3 VARIABLES
Interest | Inflation | FX

COURSE GFB3043 / GEB3043
SEMESTER SEP 2026
GROUP [XX] | STUDENTS: [FULL NAME • STUDENT ID • IT / IS / CS] × up to five
DRAFT • add real identities

01 · ISSUE & AUDIENCE

The learning problem

For: non-finance Malaysian university students planning overseas tuition or learning international finance.
Pain point: beginners reverse the MYR/USD direction, mix inflation effects with interest rates, and mistake an IRP forward for a guaranteed future spot.

02 · FUNCTIONAL WORKFLOW

How the application works

1. **Set synthetic inputs**
   MYR/USD, MY/US rates, inflation, years, USD cost
2. **Run finance engine**
   Exact PPP, IRP, Fisher and IFE calculations
3. **Explore shocks**
   Compare three cases; change one rate by ±2.5 pp
4. **Learn and check**
   Tutor feedback, quiz, audit trail, CSV/JSON

03 · CORRECT THEORY & QUOTE

Explicit CLO3 finance linkage

PPP · inflation

$S_t = S_0 \times [(1+\pi MY)/(1+\pi US)]^T$

Covered IRP · interest

$F = S_0 \times [(1+iMY)/(1+iUS)]^T$

Fisher · expected real rate

$r = (1+i)/(1+\pi expected) - 1$

IFE / UIP · expectation

Expected spot uses i gap; not a forward contract.

Quote: MYR/USD rises → MYR depreciates; falls → appreciates.

04 · SCREENSHOT REQUIRED

Working prototype view

FINBRIDGE AI | SCENARIO ENGINE

PASTE A GENUINE STREAMLIT BROWSER SCREENSHOT HERE

App source is provided; live screenshot cannot be verified in this build environment.

05 · ILLUSTRATIVE, 1 YEAR

3 independently verified numerical cases

S_0 = 4.50 MYR/USD; US i = 5%; US π = 2%; tuition = USD 12,000.

CASE	MY i	MY π	PPP | IRP
01 Baseline	**4.0%**	**3.0%**	**4.5441 | 4.4571**
02 Inflation	4.0%	6.5%	4.6985 | 4.4571
03 Interest	7.0%	3.0%	4.5441 | 4.5857

Inflation shock → PPP rises; interest shock → IRP rises.
PPP tuition: RM 54,529 → RM 56,382 (inflation shock).

Accuracy checks

18 automated tests passed; 6 independent Decimal/parity checks in the engine; CSV/JSON export preserves assumptions.

07 · PLAIN LANGUAGE

Main interpretation

Malaysia expected inflation > US inflation → relative PPP suggests MYR depreciation.

Higher MY nominal interest at unchanged US i → covered IRP raises MYR/USD forward; it does not promise tomorrow's spot.

Fisher links expected inflation to nominal vs real rates.

08 · VALUE & FEASIBILITY

Market fit & future rollout

Alternatives: lecture slides, spreadsheet calculators, generic chatbots.
Why this differs: coherent MYR/USD scenario engine, checked math, beginner feedback and misconception quiz.
Adoption: free lecturer-led pilot; later campus hosting, official data feeds, mobile testing and language support.

09 · LIMITATIONS

Responsible modelling

All rates are fictional. PPP may diverge from short-run spot. IRP assumes no arbitrage, matching-term yields and minimal frictions. IFE / UIP requires additional expectation/risk assumptions. No live prices, tested learning gains or guaranteed forecasts.

10 · TRACEABLE SOURCES

Institutional references

CFA Institute (2026), Currency Exchange Rates • cfainstitute.org/insights/.../currency-exchange-rates-understanding-equilibrium-value
Rogerson / CME (2024), Covered Interest Parity • cmegroup.com/articles/whitepapers/...
Callen / IMF (2018), Purchasing Power Parity • imf.org/en/publications/fandd/...
St. Louis Fed (2023), Adjusting for Inflation • stlouisfed.org/publications/page-one-economics/...

DRAFT FOR GROUP COMPLETION: Insert actual group number, full member names / student IDs / IT-IS-CS programmes, actual app screenshot, personal contributions, verified references and group recording before uploading.

Course deadline: 16 November 2026 (Monday), before 11:59 PM MYT · Poster A3 PDF + approximately 10-minute video MP4 / permitted link · Submitted by group leader.